

Assignment: Project Evaluation & Feasibility Study for the Jahangirnagar University Student Information System (JUSIS)

1. Objective

This assignment aims to assess your understanding of **programme management, benefits management, cost-benefit analysis, cash flow forecasting, and risk evaluation** (as covered in Chapter 3). You will act as a **programme/project analyst** tasked with determining whether the development of the JUSIS is strategically, financially, and operationally viable for Jahangirnagar University *before* any coding begins.

2. Scenario / Background

Jahangirnagar University (JU) currently manages student records using a mix of manual files and disjointed spreadsheets. The university's administration has proposed the creation of a **Jahangirnagar University Student Information System (JUSIS)** to centralize admissions, academics, examinations, and alumni data.

However, the University Grants Commission (UGC) requires a rigorous **Project Evaluation Report** to justify the allocation of a special development budget. The Vice-Chancellor has appointed your team as the **Programme Office** to evaluate this initiative. You must prepare a formal report applying the concepts from Chapter 3 to prove that this investment is worthwhile and manageable.

3. Your Task (Report Structure)

You must prepare a **formal Project Evaluation Report** structured exactly as follows. Your report must explicitly use the terminology and techniques from the provided slides (Programme Mandate, Blueprint, Benefits Management, NPV, ROI, Risk Matrix, Decision Trees, etc.).

Part A: Programme Context (Not just a Project)

Refer to slides: "Programme management", "Strategic programmes", "Programme Mandate"

1. **Programme vs. Project Justification:** Justify why JUSIS should be treated as a **strategic programme** rather than a single isolated project. Identify at least **two** sub-projects that would fall under this programme (e.g., "Examination Module" and "Admission Module").
2. **Programme Mandate:** Draft a brief **Programme Mandate** (as described in your slides) for JUSIS. Describe the new capability it will deliver and its fit with JU's organizational goal of becoming a "Digital University."
3. **Vision Statement & Blueprint:** Write a one-sentence **Vision Statement** and outline the major changes in the **Blueprint** (the changes needed to obtain the new capability, e.g., new hardware, network upgrades, re-trained staff).

Part B: Benefits Management Plan

Refer to slides: "Benefits management", "Quantifying benefits"

Identify and categorize at least **six (6) distinct benefits** that JUSIS will bring to Jahangirnagar University. You must clearly categorize them according to the three types mentioned in your lecture:

Category (from slide)	Example Benefit for JUSIS (You must create these)
A) Quantified and Valued (e.g., saves X staff hours = saves BDT Y)	(e.g., Automating transcript generation saves 3 admin staff 10 hours/week. Value = 3 staff × 10hrs × BDT 500/hr × 52 weeks = BDT 780,000/year)
B) Quantified but not Valued (e.g., 50% decrease in errors)	(Provide your own metric here)
C) Identified but not easily Quantified (e.g., improved reputation)	(Provide your own example here)

Additionally, briefly describe **who** will be responsible for monitoring these benefits after the system is deployed (i.e., how will the programme ensure the benefits are actually achieved post-launch?).

Part C: Cost-Benefit Analysis & Cash Flow Forecasting

Refer to slides: "Cost benefit analysis", "Net profit", "Pay back period", "Return on investment (ROI)"

Based on a fictitious (but realistic) 5-year projection, you are provided with the following **assumed financial data** for JUSIS:

- **Development Costs (Year 0):** BDT 4,500,000 (includes software purchase/license, customization, and hardware setup).
- **Annual Operational Costs (Years 1-5):** BDT 600,000 per year (includes hosting, maintenance, support staff).
- **Annual Benefits/Savings (Years 1-5):**
 - Year 1: BDT 800,000
 - Year 2: BDT 1,200,000
 - Year 3: BDT 1,800,000
 - Year 4: BDT 2,200,000
 - Year 5: BDT 2,600,000

Tasks:

1. Create a **cash flow table** showing: Year, Costs, Benefits, and Net Cash Flow (Benefits - Costs) for Years 0 through 5. (Note: Year 0 has no benefits).
2. Calculate the **Total Net Profit** over the 5 years.
3. Calculate the exact **Payback Period** (to the nearest month) for this project.
4. Calculate the **Return on Investment (ROI)** using the formula: $(\text{Average Annual Profit} / \text{Total Investment}) \times 100$.

Part D: Net Present Value (NPV) & Internal Rate of Return (IRR)

Refer to slides: "Net present value", "Discount factor", "Internal rate of return (IRR)"

1. **Calculate NPV:** Assuming a discount rate (interest rate) of **12%**, calculate the **Net Present Value (NPV)** for the JUSIS project.
 - *Hint:* You must apply the discount factor $1/(1+r)^t$ to the Net Cash Flow for Years 0 to 5.
 - Show your workings in a table. Based on the NPV result, state clearly: **Is this project financially worthwhile? Why or why not?**
2. **Interpret IRR:** Without performing the exact complex math, explain conceptually what the **Internal Rate of Return (IRR)** represents for this project. If the calculated IRR was 18%, and the university's borrowing rate is 12%, should the university proceed? Justify your answer using the slide's logic.

Part E: Risk Evaluation & Decision Analysis

Refer to slides: "Dealing with uncertainty", "Project risk matrix", "Decision trees"

1. **Project Risk Matrix:** Identify **four (4) major risks** to the JUSIS programme (e.g., resistance from faculty, data migration errors, cost overruns, vendor lock-in).
 - Place these four risks on a **Project Risk Matrix** (Probability: High/Med/Low vs. Impact: High/Med/Low).
 - For the highest risk on your matrix, propose a specific mitigation strategy.
2. **Decision Tree (Critical Choice):**
 - The university must decide whether to **Build** the system from scratch using local developers, or **Buy** an off-the-shelf commercial student system.
 - Draw a simple **Decision Tree** with these two options. Assign approximate financial outcomes (e.g., Build = BDT 3M cost with 80% success rate; Buy = BDT 5M cost with 95% success rate).
 - Calculate the **Expected Monetary Value (EMV)** for each branch and recommend which path the university should take.

4. Deliverables

Each group must submit the following:

1. **The Project Evaluation Report** (PDF format) containing Parts A to E. The body of the report should be **2,500 to 3,000 words** (excluding tables and appendices).
2. A **spreadsheet file (Excel/Google Sheets)** showing your clear calculations for the Cash Flow, Payback, ROI, and NPV (Part C & D) – properly formatted with formulas.
3. **Submission Format:** Upload the PDF report and the Excel file to the designated Classroom Assignment.

Marking Criteria (Rubric)

Section	Weight	Excellent (80-100%)	Satisfactory (60-79%)	Needs Improvement (<60%)
Part A: Programme Context	15%	Clearly distinguishes programme vs project; Mandate and Blueprint are specific and highly relevant to JU.	Basic definitions applied, but lacks depth.	Confuses programme with project; mandate is missing.
Part B: Benefits Management	15%	All 6 benefits are correctly categorized with realistic, quantified figures.	Benefits listed but not properly categorized or quantified.	Benefits are generic or not categorized.
Part C: Cash Flow & ROI	25%	Cash flow table is flawless; Payback & ROI calculations are 100% accurate and shown step-by-step.	Calculations have minor errors.	Calculations are missing or entirely incorrect.
Part D: NPV & IRR	20%	Discount factors applied perfectly; NPV calculated correctly; financial interpretation is insightful.	NPV calculation attempted but has formula errors.	NPV not attempted or completely wrong.
Part E: Risk & Decisions	15%	Risk matrix is comprehensive; Decision tree uses clear EMV calculations with a well-justified choice.	Risks identified but matrix is vague; Decision tree lacks numeric EMV.	Risks and decision tree are missing.
Report Quality & Presentation	10%	Professional formatting, clear headings, proper referencing of slide concepts (e.g., "as per the OGC approach").	Adequate formatting, some grammatical errors.	Messy report, difficult to read.

7. Important Notes

- **Use the Terminology:** Your slides explicitly mention "Programme Director," "Blueprint," "Cash-flow," and "Discount Factor." You must use these exact terms in your report to demonstrate understanding.
- **Critical Thinking:** In Part E, when discussing risks, remember the golden rule from your slides: *"A project may fail not through poor management but because it should never have been started"* – explain how your risk analysis helps avoid this.
- **Data Privacy:** All financial figures provided in Part C are fictional and for academic calculation purposes only.

Good luck! Remember, a well-evaluated project is already half-successful.